DSA 8202: Time Series Analysis and Forecasting

Lecture Notes

Topic 5: ARMA Models and Model Identification

Prof. Jacob Ong'ala

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1 ARMA Model Specification

1.1 General ARMA(p, q) Model

1.1.1 Introduction

An Autoregressive Moving Average (ARMA) model combines two important components of time series modelling: the **autoregressive (AR) component** and the **moving average (MA) component**.

The AR component describes how the current observation depends on previous observations, whereas the MA component describes how the current observation is influenced by current and previous random innovations.

An ARMA(p, q) process is therefore particularly useful for modelling stationary time series that exhibit both persistence in observations and dependence on previous random shocks.

1.1.2 Mathematical Specification

The general ARMA(p, q) model is expressed as

$$Y_t = c + \sum_{i=1}^{p} \phi_i Y_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}, \tag{1}$$

where

- Y_t is the observation at time t;
- c is the intercept or constant term;
- p is the order of the autoregressive component;
- q is the order of the moving average component;
- $\phi_1, \ldots, \phi_p$ are the autoregressive parameters;
- $\theta_1, \ldots, \theta_q$ are the moving average parameters;
- ε_t is the random innovation at time t.

The innovation process is assumed to be white noise:

$$\varepsilon_t \sim WN(0, \sigma_\varepsilon^2). \tag{2}$$

Thus,

$$E(\varepsilon_t) = 0, \tag{3}$$

$$\text{Var}(\varepsilon_t) = \sigma_\varepsilon^2, \tag{4}$$

$$\text{Cov}(\varepsilon_t, \varepsilon_s) = 0, \quad t \neq s. \tag{5}$$

For Gaussian ARMA models, we additionally assume

$$\varepsilon_t \overset{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2). \tag{6}$$

1.1.3 Zero-Intercept ARMA(p, q) Model

For theoretical development, we shall primarily consider the zero-intercept ARMA model, obtained by setting $c = 0$.

The resulting specification is

$$\boxed{Y_t = \sum_{i=1}^{p} \phi_i Y_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}.} \tag{7}$$

Expanding the summations gives

$$\begin{aligned} Y_t = &\phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} \\ &+ \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}. \end{aligned} \tag{8}$$

This representation separates the current observation into three contributions:

1. Dependence on previous observations;
2. The current random innovation;
3. Effects of previous innovations.

Mean of the Zero-Intercept Model Suppose that $\{Y_t\}$ is weakly stationary with constant mean

$$E(Y_t) = \mu.$$

Taking expectations of Equation (7) yields

$$E(Y_t) = \sum_{i=1}^{p} \phi_i E(Y_{t-i}) + E(\varepsilon_t) + \sum_{j=1}^{q} \theta_j E(\varepsilon_{t-j}). \tag{9}$$

Since the innovations have zero mean,

$$E(\varepsilon_t) = E(\varepsilon_{t-j}) = 0,$$

and stationarity implies

$$E(Y_t) = E(Y_{t-i}) = \mu.$$

Consequently,

$$\mu = \sum_{i=1}^{p} \phi_i \mu. \tag{10}$$

Rearranging,

$$\mu \left(1 - \sum_{i=1}^{p} \phi_i\right) = 0. \tag{11}$$

For a causal stationary ARMA model,

$$1 - \sum_{i=1}^{p} \phi_i \neq 0.$$

Therefore,

$$\boxed{E(Y_t) = \mu = 0.} \tag{12}$$

Hence, a stationary zero-intercept ARMA process has zero unconditional mean.

Relationship Between the Intercept and Mean For the general model with intercept c, taking expectations gives

$$\mu = c + \sum_{i=1}^{p} \phi_i \mu.$$

Therefore,

$$\boxed{c = \mu \left(1 - \sum_{i=1}^{p} \phi_i\right).} \tag{13}$$

This relationship shows that the intercept c and the unconditional mean μ are generally not the same quantity.

If a stationary series has a nonzero mean, we may define the centred process

$$X_t = Y_t - \mu.$$

The centred process satisfies

$$\boxed{X_t = \sum_{i=1}^{p} \phi_i X_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}.} \tag{14}$$

Thus, the zero-intercept specification can be used without loss of generality when analysing a stationary series after centring it around its mean.

1.1.4 The Autoregressive Component

The autoregressive part of the model is

$$\sum_{i=1}^{p} \phi_i Y_{t-i} = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p}. \tag{15}$$

It describes the linear contribution of previous observations to the current value.

For example, an AR(2) model is

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t. \tag{16}$$

The current observation depends on the two immediately preceding observations.

1.1.5 The Moving Average Component

The moving average component is

$$\varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}. \tag{17}$$

It describes how the current and previous innovations contribute to the observed process.

For example, an MA(2) model is

$$Y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}. \tag{18}$$

The coefficients θ_1 and θ_2 determine how shocks from the preceding two periods influence the current observation.

1.1.6 Examples of ARMA Models

Example 1: ARMA$(1,1)$ Setting $p=1$ and $q=1$ gives

$$\boxed{Y_t = \phi_1 Y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}.} \tag{19}$$

Consider the model

$$Y_t = 0.6Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}. \tag{20}$$

Suppose

$$Y_{t-1} = 10, \qquad \varepsilon_{t-1} = 2, \qquad \varepsilon_t = 1.$$

Substituting,

$$\begin{aligned} Y_t &= 0.6(10) + 1 + 0.4(2) && (21)\\ &= 6 + 1 + 0.8 && (22)\\ &= \boxed{7.8}. && (23) \end{aligned}$$

Therefore, the current observation is influenced by the previous observation, the previous innovation, and the current innovation.

Example 2: ARMA$(2,1)$ For $p=2$ and $q=1$,

$$\boxed{Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t + \theta_1 \varepsilon_{t-1}.} \tag{24}$$

This model contains two autoregressive terms and one lagged innovation term.

Example 3: ARMA$(1,2)$ For $p=1$ and $q=2$,

$$\boxed{Y_t = \phi_1 Y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}.} \tag{25}$$

The current observation depends on one previous observation and innovations from the previous two periods.

Example 4: ARMA$(2,2)$ For $p=2$ and $q=2$,

$$\boxed{\begin{aligned} Y_t &= \phi_1 Y_{t-1} + \phi_2 Y_{t-2} \\ &\quad + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}. \end{aligned}} \tag{26}$$

This model combines two autoregressive terms with two moving average terms.

1.1.7 Backshift Operator Representation

The backshift operator B is defined by

$$BY_t = Y_{t-1}. \tag{27}$$

More generally,

$$B^k Y_t = Y_{t-k}. \tag{28}$$

Define the autoregressive polynomial

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p, \tag{29}$$

and the moving average polynomial

$$\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q. \tag{30}$$

Starting from the zero-intercept model,

$$Y_t = \sum_{i=1}^{p} \phi_i Y_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}, \tag{31}$$

move the autoregressive terms to the left:

$$Y_t - \sum_{i=1}^{p} \phi_i Y_{t-i} = \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}. \tag{32}$$

Applying the backshift operator,

$$(1 - \phi_1 B - \cdots - \phi_p B^p)\, Y_t = (1 + \theta_1 B + \cdots + \theta_q B^q)\, \varepsilon_t. \tag{33}$$

Therefore, the compact ARMA representation is

$$\boxed{\phi(B) Y_t = \theta(B) \varepsilon_t.} \tag{34}$$

Example: ARMA$(1,1)$ For an ARMA$(1,1)$ process,

$$Y_t = \phi Y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}. \tag{35}$$

Using the backshift operator,

$$Y_t - \phi B Y_t = \varepsilon_t + \theta B \varepsilon_t. \tag{36}$$

Hence,

$$\boxed{(1 - \phi B) Y_t = (1 + \theta B) \varepsilon_t.} \tag{37}$$

1.1.8 Special Cases of ARMA Models

The ARMA family includes autoregressive and moving average models as special cases.

Pure Autoregressive Model When $q = 0$,

$$Y_t = \sum_{i=1}^{p} \phi_i Y_{t-i} + \varepsilon_t. \tag{38}$$

This is an AR(p) model.

Pure Moving Average Model When $p = 0$,

$$Y_t = \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}. \tag{39}$$

This is an MA(q) model.

White Noise Model When $p = q = 0$,

$$Y_t = \varepsilon_t. \tag{40}$$

This is an ARMA$(0, 0)$ process.

1.1.9 Comparison of AR, MA and ARMA Models

Table 1: Comparison of AR, MA and ARMA models

Model	Specification	Source of Dependence
AR(p)	$Y_t = \sum_{i=1}^{p} \phi_i Y_{t-i} + \varepsilon_t$	Previous observations
MA(q)	$Y_t = \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}$	Current and previous innovations
ARMA(p, q)	$\phi(B)Y_t = \theta(B)\varepsilon_t$	Previous observations and innovations

1.2 Worked Example: Calculating an ARMA Value

Consider the following ARMA$(1, 1)$ model:

$$Y_t = 4 + 0.6Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}. \tag{41}$$

This model combines an autoregressive component of order one and a moving average component of order one.

The model parameters are:

- Intercept: $c = 4$;
- Autoregressive coefficient: $\phi_1 = 0.6$;
- Moving average coefficient: $\theta_1 = 0.4$.

The unconditional mean of the process is given by

$$\mu = \frac{c}{1 - \phi_1} = \frac{4}{1 - 0.6} = \frac{4}{0.4} = 10. \tag{42}$$

Thus, the process fluctuates around a long-run mean of 10.

Step 1: Identify the Given Information

Suppose we are given the following values:

$$\begin{aligned} Y_{t-1} &= 15, \\ \varepsilon_t &= 2, \\ \varepsilon_{t-1} &= 1. \end{aligned}$$

Our objective is to calculate the current observation Y_t.

Step 2: Substitute the Values into the Model

The model is

$$Y_t = 4 + 0.6Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}.$$

Substituting the given values:

$$Y_t = 4 + 0.6(15) + 2 + 0.4(1). \tag{43}$$

Step 3: Perform the Calculations

Calculate the autoregressive contribution:

$$0.6(15) = 9.$$

Calculate the moving average contribution:

$$0.4(1) = 0.4.$$

Therefore,

$$\begin{aligned} Y_t &= 4 + 9 + 2 + 0.4 \\ &= 15.4. \end{aligned}$$

Hence, the current value of the time series is

$$\boxed{Y_t = 15.4.} \tag{44}$$

Step 4: Interpret the Result

The calculated value of 15.4 is influenced by four components:

Table 2: Contributions to the current ARMA value

Component	**Contribution**
Intercept	4.0
Autoregressive component	9.0
Current innovation	2.0
Previous innovation	0.4
Total	**15.4**

The result indicates that the current observation is above the unconditional mean of 10.

The autoregressive component carries forward information from the previous observation, while the moving average component captures the influence of the previous random shock.

2 Assumptions and Properties of ARMA Models

An Autoregressive Moving Average (ARMA) model is a powerful framework for describing the dependence structure of stationary time series. However, the validity of an ARMA model depends on several theoretical assumptions concerning the behaviour of the time series, the innovation process and the model parameters.

In this section, we examine the fundamental assumptions and properties of ARMA models, including stationarity, white-noise innovations, causality, invertibility, non-redundancy of model polynomials, linearity and parameter stability.

Throughout this section, we consider the zero-mean ARMA(p, q) model:

$$\boxed{Y_t = \sum_{i=1}^{p} \phi_i Y_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}.} \tag{45}$$

Equivalently, using the backshift operator,

$$\boxed{\phi(B)Y_t = \theta(B)\varepsilon_t,} \tag{46}$$

where

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p, \tag{47}$$
$$\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q. \tag{48}$$

For a non-zero-mean stationary process, the same development applies to the centred series $X_t = Y_t - \mu$.

2.1 Stationarity

2.1.1 General Stationarity Condition for ARMA Models

Consider the ARMA(p, q) model:

$$\phi(B)Y_t = \theta(B)\varepsilon_t.$$

The autoregressive polynomial is

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p. \tag{49}$$

The standard condition for a causal stationary ARMA process is that all roots of

$$\phi(z) = 0 \tag{50}$$

lie outside the unit circle.

That is,

$$\boxed{|z_i| > 1 \quad \text{for every root } z_i.} \tag{51}$$

This condition ensures that the autoregressive polynomial can be inverted to produce a stable infinite moving average representation.

A detailed derivation of this condition is provided in the section on causality.

2.1.2 Stationarity Condition for an ARMA$(1, 1)$ Model

Consider the zero-mean ARMA$(1, 1)$ model:

$$Y_t = \phi Y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}, \tag{52}$$

where

$$\varepsilon_t \sim WN(0, \sigma_\varepsilon^2).$$

The parameter ϕ is the autoregressive coefficient, while θ is the moving average coefficient.

Step 1: Express the Model Using the Backshift Operator Using

$$BY_t = Y_{t-1},$$

the model becomes

$$Y_t - \phi B Y_t = \varepsilon_t + \theta B \varepsilon_t.$$

Therefore,

$$\boxed{(1-\phi B)Y_t = (1+\theta B)\varepsilon_t.} \tag{53}$$

Step 2: Obtain the Infinite MA Representation Rearranging Equation (53),

$$Y_t = \frac{1+\theta B}{1-\phi B}\varepsilon_t.$$

Using the geometric series expansion,

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots, \qquad |x| < 1,$$

we obtain

$$\frac{1}{1-\phi B} = 1 + \phi B + \phi^2 B^2 + \phi^3 B^3 + \cdots.$$

Substituting,

$$Y_t = (1+\theta B)(1 + \phi B + \phi^2 B^2 + \cdots)\varepsilon_t.$$

Multiplying the polynomials gives

$$\begin{aligned} Y_t = \big[1 &+ (\phi+\theta)B + \phi(\phi+\theta)B^2 \\ &+ \phi^2(\phi+\theta)B^3 + \cdots\big]\varepsilon_t. \end{aligned}$$

Consequently,

$$\begin{aligned} Y_t = \varepsilon_t &+ (\phi+\theta)\varepsilon_{t-1} \\ &+ \phi(\phi+\theta)\varepsilon_{t-2} \\ &+ \phi^2(\phi+\theta)\varepsilon_{t-3} + \cdots. \end{aligned} \tag{54}$$

The coefficients of the infinite MA representation are

$$\psi_0 = 1 \tag{55}$$

and

$$\boxed{\psi_j = \phi^{j-1}(\phi+\theta), \qquad j \geq 1.} \tag{56}$$

Step 3: Establish the Stationarity Condition For the standard causal representation, the coefficients must be absolutely summable:

$$\sum_{j=0}^{\infty} |\psi_j| < \infty.$$

Substituting the coefficients,

$$1 + |\phi+\theta| \sum_{j=1}^{\infty} |\phi|^{j-1} < \infty.$$

Using the geometric series,

$$1 + \frac{|\phi + \theta|}{1 - |\phi|} < \infty.$$

This holds when

$$|\phi| < 1.$$

Therefore, the stationarity condition for a causal ARMA$(1,1)$ model is

$$\boxed{-1 < \phi < 1.} \tag{57}$$

The moving average coefficient θ does not determine the stationarity condition.

2.1.3 Worked Example 1: Stationary ARMA$(1,1)$

Consider the model

$$Y_t = 0.6Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}, \tag{58}$$

where

$$\varepsilon_t \sim WN(0, 4).$$

The parameters are

$$\phi = 0.6, \qquad \theta = 0.4.$$

The stationarity condition is

$$|\phi| < 1.$$

Since

$$|0.6| < 1,$$

the process is stationary and causal.

The infinite MA representation is

$$\begin{aligned} Y_t = \varepsilon_t &+ (0.6 + 0.4)\varepsilon_{t-1} \\ &+ 0.6(0.6 + 0.4)\varepsilon_{t-2} \\ &+ 0.6^2(0.6 + 0.4)\varepsilon_{t-3} + \cdots . \end{aligned}$$

Therefore,

$$\boxed{Y_t = \varepsilon_t + \varepsilon_{t-1} + 0.6\varepsilon_{t-2} + 0.36\varepsilon_{t-3} + \cdots .}$$

The unconditional variance is

$$\gamma(0) = \frac{\sigma_\varepsilon^2(1 + 2\phi\theta + \theta^2)}{1 - \phi^2}.$$

Substituting,

$$\begin{aligned} \gamma(0) &= \frac{4[1 + 2(0.6)(0.4) + (0.4)^2]}{1 - (0.6)^2} \\ &= \frac{4(1 + 0.48 + 0.16)}{0.64} \\ &= \frac{6.56}{0.64} \\ &= \boxed{10.25}. \end{aligned}$$

Thus, the process is stationary with unconditional variance 10.25.

2.1.4 Worked Example 2: Non-Stationary ARMA$(1,1)$

Consider

$$Y_t = 1.2Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}. \tag{59}$$

Here,

$$\phi = 1.2, \qquad \theta = 0.4.$$

The stationarity condition requires

$$|\phi| < 1.$$

However,

$$|1.2| > 1.$$

Therefore, the process does not have a causal stationary representation.

The effect of previous observations grows rather than decays over time.

The moving average coefficient 0.4 cannot correct this violation.

Conclusion For an ARMA$(1,1)$ model,

$$Y_t = \phi Y_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1},$$

the standard causal stationarity condition is

$$\boxed{|\phi| < 1.} \tag{60}$$

The autoregressive coefficient determines causal stationarity, while the moving average coefficient determines the invertibility condition.

2.1.5 Factors That May Violate Stationarity

Several features in observed time series may indicate non-stationarity.

1. Trend A series with a systematic upward or downward movement may have a time-varying mean.

For example,

$$Y_t = 5 + 2t + \varepsilon_t. \tag{61}$$

Taking expectations,

$$E(Y_t) = 5 + 2t.$$

The mean changes with time; hence, the series is not weakly stationary.

2. Changing Variance Consider

$$Y_t = \sigma_t \varepsilon_t,$$

where ε_t has unit variance and σ_t^2 changes over time.

Then,

$$\mathrm{Var}(Y_t) = \sigma_t^2.$$

Since the variance is time-dependent, the process is not weakly stationary.

3. Structural Breaks A structural break occurs when the statistical properties of a series change at a particular point in time.

For example,

$$Y_t = \begin{cases} 10 + \varepsilon_t, & t \leq T, \\ 20 + \varepsilon_t, & t > T. \end{cases}$$



The mean changes from 10 to 20 at time T.

4. Changing Seasonal Patterns A seasonal pattern that changes in magnitude or behaviour over time may violate stationarity.

Such patterns should be investigated before fitting a non-seasonal ARMA model.

2.2 Innovation or White-Noise Assumptions

2.2.1 Definition of White Noise

The innovations represent the random shocks that cannot be explained by the systematic component of the time series model.

A common assumption is that the innovation sequence $\{\varepsilon_t\}$ is white noise with zero mean and constant variance.

Mathematically,

$$\boxed{\varepsilon_t \sim WN(0, \sigma_\varepsilon^2).} \tag{62}$$

This means that

$$E(\varepsilon_t) = 0, \tag{63}$$
$$\text{Var}(\varepsilon_t) = \sigma_\varepsilon^2, \tag{64}$$
$$\text{Cov}(\varepsilon_t, \varepsilon_s) = 0, \qquad t \neq s. \tag{65}$$

The covariance condition means that innovations at different time periods are uncorrelated.

White noise does not necessarily imply independence. Independence is a stronger condition.

2.2.2 Gaussian White Noise

For Gaussian ARMA modelling, we may additionally assume

$$\boxed{\varepsilon_t \overset{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2).} \tag{66}$$

This assumption implies that the innovations are normally distributed and independent over time.

Normality is not required for the basic definition of an ARMA process.

However, it is useful for constructing the Gaussian likelihood function and conducting statistical inference.

Under Gaussian assumptions, uncorrelated innovations are also independent.

2.2.3 Gaussian Likelihood

Suppose that

$$\varepsilon_t \sim N(0, \sigma_\varepsilon^2).$$

The density of an individual innovation is

$$f(\varepsilon_t) = \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \exp\left(-\frac{\varepsilon_t^2}{2\sigma_\varepsilon^2}\right). \tag{67}$$

Assuming independent Gaussian innovations, the joint likelihood is proportional to

$$L(\boldsymbol{\eta}, \sigma_\varepsilon^2) = \prod_{t=1}^{n} \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \exp\left(-\frac{\varepsilon_t^2}{2\sigma_\varepsilon^2}\right), \tag{68}$$

where $\boldsymbol{\eta}$ denotes the ARMA parameters.

Taking logarithms,

$$\ell = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon^2}\sum_{t=1}^{n}\varepsilon_t^2. \tag{69}$$

For fixed ARMA parameters, differentiating with respect to σ_ε^2 gives

$$\frac{\partial \ell}{\partial \sigma_\varepsilon^2} = -\frac{n}{2\sigma_\varepsilon^2} + \frac{1}{2(\sigma_\varepsilon^2)^2}\sum_{t=1}^{n}\varepsilon_t^2. \tag{70}$$

Setting the derivative equal to zero,

$$-\frac{n}{2\sigma_\varepsilon^2} + \frac{\sum_{t=1}^{n}\varepsilon_t^2}{2(\sigma_\varepsilon^2)^2} = 0.$$

Multiplying through and rearranging,

$$\boxed{\widehat{\sigma}_\varepsilon^2 = \frac{1}{n}\sum_{t=1}^{n}\widehat{\varepsilon}_t^2.} \tag{71}$$

This is the Gaussian maximum likelihood variance estimate when the innovations are observed or replaced by their fitted residuals under the corresponding likelihood formulation.

For exact ARMA likelihood estimation, the initial observations and their conditional distribution must also be handled appropriately.

2.3 Causality

2.3.1 Meaning of Causality

An ARMA process is causal if its current value can be expressed as a convergent linear combination of current and past innovations.

This means that the current observation depends only on present and past random shocks, rather than future shocks.

The causal representation is

$$\boxed{Y_t - \mu = \sum_{j=0}^{\infty}\psi_j\varepsilon_{t-j}.} \tag{72}$$

The coefficients $\psi_0, \psi_1, \psi_2, \ldots$ describe how the effect of an innovation propagates through time.

For a standard causal representation, we require

$$\boxed{\sum_{j=0}^{\infty}|\psi_j| < \infty.} \tag{73}$$

This condition ensures absolute summability of the impulse-response coefficients.

2.3.2 Derivation of the Causality Condition

Consider the zero-mean ARMA model:

$$\phi(B)Y_t = \theta(B)\varepsilon_t.$$

If the autoregressive polynomial can be inverted, we obtain

$$Y_t = \frac{\theta(B)}{\phi(B)}\varepsilon_t. \tag{74}$$

Suppose that

$$\frac{\theta(B)}{\phi(B)} = \sum_{j=0}^{\infty} \psi_j B^j.$$

Then

$$Y_t = \sum_{j=0}^{\infty} \psi_j B^j \varepsilon_t. \tag{75}$$

Using the backshift operator,

$$B^j \varepsilon_t = \varepsilon_{t-j},$$

we obtain

$$\boxed{Y_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}.} \tag{76}$$

Therefore, causality depends on whether the reciprocal of the autoregressive polynomial has a convergent power-series expansion.

2.3.3 Derivation of the Causality Condition for an ARMA$(1,1)$ Model

Consider the zero-mean ARMA$(1,1)$ model:

$$\boxed{Y_t = \phi Y_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1}.} \tag{77}$$

Our objective is to establish the condition under which the process can be expressed as a convergent infinite moving average of current and past innovations.

Step 1: Express the Model Using the Backshift Operator Recall that

$$BY_t = Y_{t-1}$$

and

$$B\varepsilon_t = \varepsilon_{t-1}.$$

The ARMA$(1,1)$ model becomes

$$Y_t = \phi BY_t + \varepsilon_t + \theta B\varepsilon_t.$$

Collecting the terms involving Y_t,

$$Y_t - \phi BY_t = (1 + \theta B)\varepsilon_t.$$

Therefore,

$$\boxed{(1 - \phi B)Y_t = (1 + \theta B)\varepsilon_t.} \tag{78}$$

Step 2: Express Y_t in Terms of Innovations Rearranging,

$$Y_t = \frac{1+\theta B}{1-\phi B}\varepsilon_t.$$

We can write this as

$$Y_t = (1+\theta B)\frac{1}{1-\phi B}\varepsilon_t.$$

To obtain the infinite moving average representation, we need to expand the reciprocal term.

Step 3: Apply the Geometric Series Expansion Recall the geometric series:

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots, \qquad |x| < 1.$$

Taking $x = \phi B$, we obtain

$$\frac{1}{1-\phi B} = 1 + \phi B + \phi^2 B^2 + \phi^3 B^3 + \cdots.$$

This expansion is valid under the condition

$$|\phi| < 1.$$

Substituting into the ARMA model,

$$Y_t = (1+\theta B)(1 + \phi B + \phi^2 B^2 + \phi^3 B^3 + \cdots)\varepsilon_t.$$

Step 4: Multiply the Polynomials Expanding the expression,

$$\begin{aligned} Y_t = &\left(1 + \phi B + \phi^2 B^2 + \phi^3 B^3 + \cdots\right)\varepsilon_t \\ &+ \theta B\left(1 + \phi B + \phi^2 B^2 + \phi^3 B^3 + \cdots\right)\varepsilon_t. \end{aligned}$$

Multiplying each term gives

$$\begin{aligned} Y_t = &\varepsilon_t + \phi B\varepsilon_t + \phi^2 B^2\varepsilon_t + \phi^3 B^3\varepsilon_t + \cdots \\ &+ \theta B\varepsilon_t + \theta\phi B^2\varepsilon_t + \theta\phi^2 B^3\varepsilon_t + \cdots. \end{aligned}$$

Collecting terms with the same powers of B,

$$\begin{aligned} Y_t = &\varepsilon_t + (\phi+\theta)B\varepsilon_t \\ &+ \left(\phi^2 + \theta\phi\right)B^2\varepsilon_t \\ &+ \left(\phi^3 + \theta\phi^2\right)B^3\varepsilon_t + \cdots. \end{aligned}$$

Factorising the coefficients,

$$\begin{aligned} Y_t = &\varepsilon_t + (\phi+\theta)B\varepsilon_t \\ &+ \phi(\phi+\theta)B^2\varepsilon_t \\ &+ \phi^2(\phi+\theta)B^3\varepsilon_t + \cdots. \end{aligned}$$

Using the backshift operator,

$$B^j\varepsilon_t = \varepsilon_{t-j},$$

we obtain

$$\boxed{\begin{aligned} Y_t = &\varepsilon_t + (\phi+\theta)\varepsilon_{t-1} \\ &+ \phi(\phi+\theta)\varepsilon_{t-2} \\ &+ \phi^2(\phi+\theta)\varepsilon_{t-3} + \cdots. \end{aligned}} \tag{79}$$

This is the infinite MA representation of the ARMA$(1,1)$ process.

Step 5: Identify the Infinite MA Coefficients Recall the general causal representation:

$$Y_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}.$$

Comparing this with the derived representation, we obtain

$$\psi_0 = 1, \tag{80}$$
$$\psi_1 = \phi + \theta, \tag{81}$$
$$\psi_2 = \phi(\phi + \theta), \tag{82}$$
$$\psi_3 = \phi^2(\phi + \theta). \tag{83}$$

Therefore, the general coefficient is

$$\boxed{\psi_j = \phi^{j-1}(\phi + \theta), \qquad j \geq 1.} \tag{84}$$

Step 6: Establish the Causality Condition For the process to be causal, the infinite MA coefficients must be absolutely summable:

$$\sum_{j=0}^{\infty} |\psi_j| < \infty.$$

Substituting the derived coefficients,

$$1 + \sum_{j=1}^{\infty} \left|\phi^{j-1}(\phi + \theta)\right| < \infty.$$

Taking constants outside the summation,

$$1 + |\phi + \theta| \sum_{j=1}^{\infty} |\phi|^{j-1} < \infty.$$

Using the geometric series,

$$\sum_{j=1}^{\infty} |\phi|^{j-1} = 1 + |\phi| + |\phi|^2 + \cdots = \frac{1}{1 - |\phi|},$$

provided $|\phi| < 1$.

Thus,

$$1 + \frac{|\phi + \theta|}{1 - |\phi|} < \infty.$$

Consequently, the required condition is

$$\boxed{|\phi| < 1.} \tag{85}$$

Therefore, the ARMA$(1, 1)$ model is causal if

$$\boxed{-1 < \phi < 1.} \tag{86}$$

Notice that the moving average coefficient θ does not determine causality.

Step 7: Alternative Derivation Using the Characteristic Polynomial The autoregressive polynomial for ARMA$(1,1)$ is

$$\phi(z) = 1 - \phi z.$$

To determine the causality condition, solve

$$1 - \phi z = 0.$$

Therefore,

$$z = \frac{1}{\phi}.$$

The causality condition requires the root to lie outside the unit circle:

$$|z| > 1.$$

Substituting the root,

$$\left|\frac{1}{\phi}\right| > 1.$$

Hence,

$$\boxed{|\phi| < 1.} \tag{87}$$

This confirms the result obtained using the infinite MA representation.

2.3.4 Worked Example: Causality of an ARMA$(1,1)$ Model

Consider the ARMA$(1,1)$ model

$$Y_t = 0.6Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}. \tag{88}$$

Here,

$$\phi = 0.6, \qquad \theta = 0.4.$$

The infinite MA representation is

$$\begin{aligned} Y_t = \varepsilon_t &+ (0.6 + 0.4)\varepsilon_{t-1} \\ &+ 0.6(0.6 + 0.4)\varepsilon_{t-2} \\ &+ 0.6^2(0.6 + 0.4)\varepsilon_{t-3} + \cdots. \end{aligned}$$

Simplifying,

$$\boxed{Y_t = \varepsilon_t + \varepsilon_{t-1} + 0.6\varepsilon_{t-2} + 0.36\varepsilon_{t-3} + 0.216\varepsilon_{t-4} + \cdots.}$$

The coefficients are

$$\psi_0 = 1, \quad \psi_1 = 1, \quad \psi_2 = 0.6, \quad \psi_3 = 0.36, \quad \psi_4 = 0.216.$$

Their absolute sum is

$$\begin{aligned} \sum_{j=0}^{\infty} |\psi_j| &= 1 + 1 + 0.6 + 0.36 + 0.216 + \cdots \\ &= 1 + \frac{1}{1 - 0.6} \\ &= 1 + 2.5 \\ &= \boxed{3.5}. \end{aligned}$$

Since the sum is finite, the process is causal.

Conclusion For an ARMA(1, 1) model,

$$Y_t = \phi Y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1},$$

the standard causality condition is

$$\boxed{|\phi| < 1.} \tag{89}$$

The autoregressive coefficient determines causality, whereas the moving average coefficient does not affect this condition.

2.4 Invertibility

2.4.1 Meaning of Invertibility

An ARMA process is invertible if the innovations can be expressed as a convergent linear combination of current and past observations.

In other words, invertibility allows us to recover the random shocks from the observed time series.

The invertible representation is

$$\boxed{\varepsilon_t = \sum_{j=0}^{\infty} \pi_j (Y_{t-j} - \mu).} \tag{90}$$

For a zero-mean process, this becomes

$$\varepsilon_t = \sum_{j=0}^{\infty} \pi_j Y_{t-j}.$$

We require

$$\boxed{\sum_{j=0}^{\infty} |\pi_j| < \infty.} \tag{91}$$

Invertibility is important because it ensures that the innovations can be recovered in a stable and unique way from the observed series.

2.4.2 Derivation of the Invertibility Condition for ARMA(1,1)

Consider the ARMA(1,1) model given by

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}, \tag{92}$$

where c is the intercept, ϕ is the autoregressive parameter, θ is the moving average parameter, and ε_t is white noise with

$$E(\varepsilon_t) = 0, \qquad \text{Var}(\varepsilon_t) = \sigma_\varepsilon^2.$$

The objective is to establish the condition under which the innovations ε_t can be expressed as a convergent infinite linear combination of the current and past observations.

Step 1: Express the model using the backshift operator Recall that the backshift operator is defined by

$$BY_t = Y_{t-1}, \qquad B\varepsilon_t = \varepsilon_{t-1}.$$

The ARMA(1,1) model can therefore be written as

$$Y_t = c + \phi B Y_t + \varepsilon_t + \theta B \varepsilon_t. \tag{93}$$

Rearranging gives

$$(1 - \phi B)Y_t = c + (1 + \theta B)\varepsilon_t. \tag{94}$$

For a stationary process, the unconditional mean is

$$\mu = E(Y_t) = \frac{c}{1 - \phi},$$

provided that $\phi \neq 1$ and the process is stationary.

Subtracting the mean from both sides gives the mean-corrected representation

$$\boxed{(1 - \phi B)(Y_t - \mu) = (1 + \theta B)\varepsilon_t.} \tag{95}$$

Step 2: Express the innovations in terms of observations Starting from

$$(1 - \phi B)(Y_t - \mu) = (1 + \theta B)\varepsilon_t,$$

divide both sides by the moving average polynomial $(1 + \theta B)$ to obtain

$$\varepsilon_t = \frac{1 - \phi B}{1 + \theta B}(Y_t - \mu). \tag{96}$$

Thus, the innovations can be expressed as

$$\boxed{\varepsilon_t = (1 - \phi B)(1 + \theta B)^{-1}(Y_t - \mu).} \tag{97}$$

Step 3: Expand the inverse moving average polynomial Using the geometric series expansion,

$$\frac{1}{1 + x} = 1 - x + x^2 - x^3 + \cdots, \qquad |x| < 1,$$

we obtain

$$\frac{1}{1 + \theta B} = 1 - \theta B + \theta^2 B^2 - \theta^3 B^3 + \cdots. \tag{98}$$

This expansion is valid when

$$\boxed{|\theta| < 1.} \tag{99}$$

Substituting into the expression for ε_t gives

$$\begin{aligned} \varepsilon_t = (1 - \phi B)(1 - \theta B + \theta^2 B^2 - \theta^3 B^3 + \cdots) \\ \times (Y_t - \mu). \end{aligned} \tag{100}$$

Step 4: Multiply and collect terms Expanding the two polynomials gives

$$\begin{aligned} \varepsilon_t = \big[1 - \theta B + \theta^2 B^2 - \theta^3 B^3 + \cdots \\ -\phi B + \phi\theta B^2 - \phi\theta^2 B^3 + \cdots\big](Y_t - \mu). \end{aligned}$$

Collecting terms involving the same powers of B,

$$\begin{aligned} \varepsilon_t = \big[1 - (\phi + \theta)B + (\theta^2 + \phi\theta)B^2 \\ -(\theta^3 + \phi\theta^2)B^3 + \cdots\big](Y_t - \mu). \end{aligned}$$

Factorising the coefficients,

$$\begin{aligned}\varepsilon_t = \big[&1-(\phi+\theta)B+\theta(\phi+\theta)B^2\\ &-\theta^2(\phi+\theta)B^3+\theta^3(\phi+\theta)B^4-\cdots\big](Y_t-\mu).\end{aligned}$$

Replacing the backshift operator with its time-series interpretation, we obtain

$$\boxed{\begin{aligned}\varepsilon_t = (Y_t-\mu)&-(\phi+\theta)(Y_{t-1}-\mu)\\ &+\theta(\phi+\theta)(Y_{t-2}-\mu)\\ &-\theta^2(\phi+\theta)(Y_{t-3}-\mu)\\ &+\theta^3(\phi+\theta)(Y_{t-4}-\mu)+\cdots.\end{aligned}} \tag{101}$$

This expression shows that the current innovation can be reconstructed from the current and past observations.

Step 5: Identify the invertibility coefficients The general invertible representation of a stationary time series is

$$\varepsilon_t = \sum_{j=0}^{\infty}\pi_j(Y_{t-j}-\mu), \tag{102}$$

where π_j are the invertibility coefficients.

Comparing this expression with the expansion obtained above, we identify

$$\pi_0 = 1,$$

and, for $j \geq 1$,

$$\begin{aligned}\pi_1 &= -(\phi+\theta),\\ \pi_2 &= \theta(\phi+\theta),\\ \pi_3 &= -\theta^2(\phi+\theta),\\ \pi_4 &= \theta^3(\phi+\theta).\end{aligned}$$

The general expression is therefore

$$\boxed{\pi_j = -(\phi+\theta)(-\theta)^{j-1}, \qquad j \geq 1.} \tag{103}$$

Step 6: Establish the convergence condition For the infinite representation to be stable and absolutely summable, we require

$$\sum_{j=0}^{\infty}|\pi_j| < \infty. \tag{104}$$

Substituting the expressions for the coefficients,

$$\begin{aligned}\sum_{j=0}^{\infty}|\pi_j| &= 1+\sum_{j=1}^{\infty}|-(\phi+\theta)(-\theta)^{j-1}|\\ &= 1+|\phi+\theta|\sum_{j=1}^{\infty}|\theta|^{j-1}\\ &= 1+|\phi+\theta|(1+|\theta|+|\theta|^2+\cdots).\end{aligned}$$

Using the sum of an infinite geometric series,

$$1 + r + r^2 + r^3 + \cdots = \frac{1}{1-r}, \qquad |r| < 1,$$

we obtain

$$\sum_{j=0}^{\infty} |\pi_j| = 1 + \frac{|\phi + \theta|}{1 - |\theta|}. \tag{105}$$

This sum is finite if

$$|\theta| < 1.$$

Consequently, the invertibility condition for the ARMA(1,1) model is

$$\boxed{|\theta| < 1.} \tag{106}$$

Notice that the invertibility condition depends on the moving average parameter θ and not on the autoregressive parameter ϕ.

2.4.3 Worked Example: Testing Invertibility of an ARMA(1,1) Model

Consider the ARMA(1,1) model

$$Y_t = 4 + 0.6Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}. \tag{107}$$

Determine whether the process is invertible and derive its infinite innovation representation.

Solution Comparing the given model with the general ARMA(1,1) model,

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1},$$

we identify

$$c = 4, \qquad \phi = 0.6, \qquad \theta = 0.4.$$

Step 1: Determine the unconditional mean

The unconditional mean is

$$\mu = \frac{c}{1-\phi}.$$

Substituting the parameter values,

$$\mu = \frac{4}{1-0.6} = \frac{4}{0.4} = 10.$$

Therefore, the centered model is

$$Y_t - 10 = 0.6(Y_{t-1} - 10) + \varepsilon_t + 0.4\varepsilon_{t-1}.$$

Step 2: Determine the root of the MA polynomial

The moving average polynomial is

$$\theta(z) = 1 + 0.4z.$$

Setting it equal to zero,

$$1 + 0.4z = 0.$$

Therefore,

$$z = -\frac{1}{0.4} = -2.5.$$

Since

$$|z| = 2.5 > 1,$$

the root lies outside the unit circle.

Hence, the process is invertible.

Step 3: Derive the infinite innovation representation

Starting from

$$\varepsilon_t = \frac{1 - 0.6B}{1 + 0.4B}(Y_t - 10),$$

we use

$$\frac{1}{1 + 0.4B} = 1 - 0.4B + 0.16B^2 - 0.064B^3 + \cdots .$$

Therefore,

$$\begin{aligned}\varepsilon_t = (1 - 0.6B)(1 - 0.4B + 0.16B^2 \\ - 0.064B^3 + \cdots)(Y_t - 10).\end{aligned}$$

Multiplying and collecting terms,

$$\begin{aligned}\varepsilon_t = (1 - B + 0.4B^2 - 0.16B^3 \\ + 0.064B^4 - \cdots)(Y_t - 10).\end{aligned}$$

Thus,

$$\boxed{\begin{aligned}\varepsilon_t = (Y_t - 10) - (Y_{t-1} - 10) \\ + 0.4(Y_{t-2} - 10) \\ - 0.16(Y_{t-3} - 10) \\ + 0.064(Y_{t-4} - 10) + \cdots .\end{aligned}} \tag{108}$$

The coefficients decrease geometrically because $|\theta| = 0.4 < 1$. Therefore, the infinite series converges absolutely.

Conclusion The ARMA(1,1) model

$$Y_t = 4 + 0.6Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}$$

is invertible because the root of its moving average polynomial lies outside the unit circle.

The general invertibility condition for an ARMA(1,1) model is

$$\boxed{|\theta| < 1.} \tag{109}$$

Important distinction between causality and invertibility For an ARMA(1,1) model,

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1},$$

the two properties are determined by different polynomials:

- **Causality:** Determined by the autoregressive polynomial $1 - \phi z$, with condition

$$\boxed{|\phi| < 1.}$$

- **Invertibility:** Determined by the moving average polynomial $1 + \theta z$, with condition

$$\boxed{|\theta| < 1.}$$

These are the standard conditions for a minimal ARMA representation, in which the autoregressive and moving average polynomials have no common roots.

2.4.4 Difference Between Causality and Invertibility

Causality and invertibility are related but distinct properties.

Property	**Causality**	**Invertibility**
Purpose	Expresses observations using past innovations	Expresses innovations using past observations
Polynomial	Autoregressive polynomial $\phi(z)$	Moving average polynomial $\theta(z)$
Condition	AR roots outside unit circle	MA roots outside unit circle
Representation	$Y_t - \mu = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$	$\varepsilon_t = \sum_{j=0}^{\infty} \pi_j (Y_{t-j} - \mu)$

An ARMA model may be causal but not invertible, invertible but not causal, both, or neither.

2.5 A Comprehensive Worked Example: Checking ARMA Model Properties

Consider the following zero-mean ARMA(1,1) model:

$$\boxed{Y_t = 0.6Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1},} \tag{110}$$

where

$$\varepsilon_t \sim WN(0, 4).$$

We want to determine whether the model is causal, stationary and invertible, and calculate its infinite moving average representation.

Step 1: Identify the Model Parameters

The model is an ARMA(1,1) process with

$$\phi_1 = 0.6, \qquad \theta_1 = 0.4.$$

The innovation variance is

$$\sigma_\varepsilon^2 = 4.$$

The autoregressive and moving average polynomials are

$$\phi(B) = 1 - 0.6B, \tag{111}$$
$$\theta(B) = 1 + 0.4B. \tag{112}$$

Step 2: Check Stationarity and Causality

The autoregressive polynomial is

$$\phi(z) = 1 - 0.6z.$$

Setting it equal to zero,

$$1 - 0.6z = 0.$$

Therefore,

$$z = \frac{1}{0.6} = 1.667.$$

Since

$$|1.667| > 1,$$

the root lies outside the unit circle.

Hence, the process is causal and has a stationary causal representation.

Step 3: Derive the Infinite MA Representation

Starting with

$$(1 - 0.6B)Y_t = (1 + 0.4B)\varepsilon_t,$$

we obtain

$$Y_t = \frac{1 + 0.4B}{1 - 0.6B}\varepsilon_t.$$

Using the geometric series,

$$\frac{1}{1 - 0.6B} = 1 + 0.6B + 0.6^2B^2 + 0.6^3B^3 + \cdots .$$

Therefore,

$$\begin{aligned} Y_t = (1 + 0.4B)(1 + 0.6B + 0.36B^2 \\ + 0.216B^3 + \cdots)\varepsilon_t. \end{aligned}$$

Multiplying the polynomials,

$$\begin{aligned} Y_t = (1 + 1.0B + 0.60B^2 \\ + 0.36B^3 + 0.216B^4 + \cdots)\varepsilon_t. \end{aligned}$$

Thus,

$$\begin{aligned} Y_t = \varepsilon_t + 1.0\varepsilon_{t-1} + 0.60\varepsilon_{t-2} \\ + 0.36\varepsilon_{t-3} + 0.216\varepsilon_{t-4} + \cdots . \end{aligned} \tag{113}$$

The coefficients are

$$\boxed{\psi_0 = 1, \qquad \psi_j = (0.6 + 0.4)(0.6)^{j-1} = (0.6)^{j-1}, \quad j \geq 1.} \tag{114}$$

Their absolute sum is

$$\sum_{j=0}^{\infty} |\psi_j| = 1 + \sum_{j=1}^{\infty} (0.6)^{j-1} \tag{115}$$

$$= 1 + \frac{1}{1 - 0.6} \tag{116}$$

$$= 1 + 2.5 \tag{117}$$

$$= \boxed{3.5}. \tag{118}$$

Since the sum is finite, the process is causal.

Step 4: Check Invertibility

The moving average polynomial is

$$\theta(z) = 1 + 0.4z.$$

Setting it equal to zero,

$$1 + 0.4z = 0.$$

Therefore,

$$z = -2.5.$$

Since

$$|-2.5| > 1,$$

the root lies outside the unit circle.

Hence,

$$\boxed{\text{The process is invertible.}}$$

Step 5: Calculate the Unconditional Mean

Taking expectations of the model,

$$E(Y_t) = 0.6E(Y_{t-1}) + E(\varepsilon_t) + 0.4E(\varepsilon_{t-1}).$$

Since the innovations have zero mean,

$$\mu = 0.6\mu.$$

Thus,

$$0.4\mu = 0,$$

and

$$\boxed{\mu = 0.}$$

Step 6: Calculate the Unconditional Variance

Using the infinite MA representation,

$$Y_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j},$$

and the fact that the innovations are uncorrelated,

$$\text{Var}(Y_t) = \sigma_\varepsilon^2 \sum_{j=0}^{\infty} \psi_j^2. \tag{119}$$

For this example,

$$\psi_0 = 1, \qquad \psi_j = (0.6)^{j-1}, \quad j \geq 1.$$

Therefore,

$$\gamma(0) = 4\left[1 + \sum_{j=1}^{\infty}(0.6)^{2(j-1)}\right] \tag{120}$$

$$= 4\left[1 + \sum_{j=0}^{\infty}(0.36)^{j}\right] \tag{121}$$

$$= 4\left[1 + \frac{1}{1-0.36}\right] \tag{122}$$

$$= 4\left(1 + \frac{1}{0.64}\right) \tag{123}$$

$$= 4(2.5625) \tag{124}$$

$$= \boxed{10.25}. \tag{125}$$

Thus, the unconditional variance is 10.25.

Step 7: Check for Common Factors

The polynomials are

$$\phi(z) = 1 - 0.6z$$

and

$$\theta(z) = 1 + 0.4z.$$

Their roots are

$$z_\phi = 1.667, \qquad z_\theta = -2.5.$$

Since the roots are different, there is no common factor.

Therefore, the model has no redundant common AR and MA factor.

Step 8: Final Conclusion

The ARMA(1,1) model satisfies the standard conditions for a causal, stationary and invertible representation.

Its main properties are summarised below:

Property	**Result**
AR coefficient	0.6
MA coefficient	0.4
Innovation variance	4
Unconditional mean	0
Unconditional variance	10.25
AR polynomial root	1.667
MA polynomial root	-2.5
Causality	Satisfied
Stationary causal representation	Satisfied
Invertibility	Satisfied
Common factors	None

3 Identification Using ACF and PACF

3.1 Autocorrelation function

The autocovariance at lag k is

$$\gamma(k) = \mathrm{Cov}(Y_t, Y_{t-k}),$$

and the autocorrelation function is

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)}.$$

The ACF measures the linear association between observations separated by k time periods.

3.2 Partial autocorrelation function

The partial autocorrelation at lag k measures the association between Y_t and Y_{t-k} after accounting for the intervening observations $Y_{t-1}, \ldots, Y_{t-k+1}$. It helps identify the autoregressive order.

3.3 Theoretical identification patterns

Model	ACF pattern	PACF pattern
AR(p)	Tails off, often exponentially or as a damped wave	Cuts off after lag p
MA(q)	Cuts off after lag q	Tails off
ARMA(p, q)	Generally tails off	Generally tails off

For a pure AR process, the PACF has a theoretical cut-off after p. For a pure MA process, the ACF has a theoretical cut-off after q. For mixed ARMA processes, both usually tail off, making identification less direct.

3.4 A practical identification procedure

1. Plot the time series and check for trend, changing variance, outliers and breaks.
2. Examine whether the series is plausibly stationary. Transform or difference only when justified.
3. Plot the sample ACF and PACF.
4. Look for an AR-like PACF cut-off, an MA-like ACF cut-off, or a mixed pattern.
5. Propose a small set of plausible low-order models, such as ARMA$(1,1)$, ARMA$(1,2)$, ARMA$(2,1)$ and ARMA$(2,2)$.
6. Estimate all candidates on the same data and compare information criteria and residual diagnostics.
7. Prefer a parsimonious model with adequate diagnostics and useful forecast performance.

3.5 Important cautions

ACF and PACF plots suggest candidate models; they do not prove the true order. Sample size, outliers, near-unit roots, parameter cancellation and model misspecification can make patterns unclear. ACF/PACF identification should therefore be combined with estimation, diagnostics and forecast evaluation.

4 Estimation of ARMA Parameters

4.1 The estimation problem

For an ARMA(p, q) model, the unknown parameters commonly include

$$\boldsymbol{\eta} = (c, \phi_1, \ldots, \phi_p, \theta_1, \ldots, \theta_q, \sigma_\varepsilon^2).$$

The objective is to estimate these quantities from the observed series. Unlike ordinary regression, lagged innovations are unobserved, so they must be reconstructed recursively or integrated out in the likelihood.

4.2 Conditional least squares

Conditional least squares (CLS) chooses parameters to minimise the sum of squared one-step-ahead residuals. For a proposed parameter vector, calculate recursively

$$\widehat{\varepsilon}_t = Y_t - \widehat{c} - \sum_{i=1}^{p} \widehat{\phi}_i Y_{t-i} - \sum_{j=1}^{q} \widehat{\theta}_j \widehat{\varepsilon}_{t-j}.$$

Then minimise

$$S(\boldsymbol{\eta}) = \sum_{t=m+1}^{n} \widehat{\varepsilon}_t^2, \qquad m = \max(p, q),$$

subject to suitable parameter constraints. Initial residuals must be specified or conditioned upon; common implementations initialise them at zero or use another prescribed initialisation.

Advantages

- Conceptually straightforward.
- Often computationally convenient.
- Can provide useful starting values for likelihood optimisation.

Limitations

- Estimates can depend on how initial residuals are treated.
- It does not fully account for the probability distribution of initial observations.
- Small-sample performance may be less efficient than maximum likelihood.

4.3 Maximum likelihood estimation

Maximum likelihood estimation (MLE) chooses parameters that make the observed data most likely under the assumed model. Under Gaussian innovations, the likelihood is based on the joint distribution of the observations.

In a conditional Gaussian likelihood, initial observations are treated as given and the residual sum of squares contributes approximately as

$$\ell_c(\boldsymbol{\eta}) = -\frac{n-m}{2} \log(2\pi\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon^2} \sum_{t=m+1}^{n} \widehat{\varepsilon}_t^2.$$

The exact likelihood accounts for the distribution and covariance structure of the initial observations, often using a state-space or innovations representation.

The MLE is

$$\widehat{\boldsymbol{\eta}}_{\text{MLE}} = \arg\max_{\boldsymbol{\eta}} \ell(\boldsymbol{\eta}).$$

Numerical optimisation is generally required because ARMA likelihoods are nonlinear in the parameters.

Advantages

- Uses a probability model for the observed series.
- Can account for initial-state uncertainty through exact likelihood methods.
- Has strong large-sample properties under appropriate regularity conditions and correct specification.
- Provides standard errors and likelihood-based tests.

Limitations

- Can be computationally demanding.
- May be sensitive to starting values or local optimisation issues.
- Gaussian likelihood assumptions may be inappropriate for heavy-tailed or strongly non-normal innovations.

4.4 CLS and MLE compared

Feature	Conditional least squares	Maximum likelihood
Criterion	Minimise conditional squared errors	Maximise likelihood
Initial values	Usually conditioned on or initialised	Can be treated through exact likelihood
Distributional assumptions	Least-squares criterion; fewer explicit distributional requirements	Requires a specified likelihood; Gaussian is common
Computation	Often simpler	Often more involved
Inference	Depends on implementation	Likelihood-based standard errors and tests

5 Model Selection and Comparison

5.1 Why compare competing models?

More than one ARMA specification may describe the observed dependence reasonably well. A model with many parameters may fit the sample closely but overfit random variation. Model selection balances goodness of fit against complexity.

5.2 Akaike Information Criterion

The Akaike Information Criterion is

$$AIC = -2\ell(\widehat{\boldsymbol{\eta}}) + 2k,$$

where ℓ is the maximised log-likelihood and k is the number of estimated parameters. Smaller AIC values indicate a better trade-off between fit and complexity among models fitted to the same data using comparable likelihoods.

5.3 Corrected AIC

For relatively small samples, the corrected AIC is

$$AIC_c = AIC + \frac{2k(k+1)}{n-k-1},$$

provided $n > k+1$. It penalises model complexity more strongly when the sample is small.

5.4 Bayesian Information Criterion

The Bayesian Information Criterion is

$$BIC = -2\ell(\widehat{\boldsymbol{\eta}}) + k\log(n).$$

BIC generally imposes a stronger penalty for additional parameters than AIC when n is sufficiently large. Smaller BIC is preferred.

5.5 Interpreting information criteria

AIC and BIC are relative measures. They do not establish that a model is absolutely correct. They should be compared for models fitted to the same response series, same observations and comparable likelihood definitions. A difference of a few units may be weak evidence; a larger difference provides stronger relative support, but practical judgement and diagnostics remain necessary.

5.6 Residual diagnostic checks

After fitting an ARMA model, define residuals

$$\widehat{\varepsilon}_t = Y_t - \widehat{Y}_{t|t-1}.$$

A satisfactory model should leave residuals that behave approximately like white noise.

Check:

1. **Residual time plot:** no obvious trend, cycles, clusters or changing variance.
2. **Residual mean:** approximately zero when an intercept or mean is included.
3. **Residual ACF:** no systematic significant autocorrelation.
4. **Ljung–Box test:** tests the joint null that a group of residual autocorrelations is zero.
5. **Distribution:** inspect a histogram and Q–Q plot if Gaussian inference or prediction intervals are important.
6. **Variance stability:** look for volatility clustering or changing residual spread.

The Ljung–Box statistic is

$$Q = n(n+2)\sum_{k=1}^{h}\frac{\widehat{\rho}_k^2}{n-k},$$

where $\widehat{\rho}_k$ is the residual sample autocorrelation at lag k and h is the maximum lag. Under suitable conditions, it is compared approximately with a chi-square distribution. When testing fitted ARMA

residuals, degrees of freedom are commonly adjusted by subtracting the number of estimated AR and MA coefficients; software implementations may differ in their conventions.

A large p-value means there is insufficient evidence to reject the white-noise null at the chosen significance level. It does not prove that residuals are independent or that the model is correct.

6 Practical Python: Fitting and Comparing ARMA Models

6.1 Software preparation

The examples use Python's `statsmodels` library. Install the packages if necessary:

```
pip install numpy pandas matplotlib statsmodels scipy
```

6.2 Simulating an ARMA(1,1) series

The following code generates a stationary ARMA(1,1) series. The chosen AR coefficient satisfies $|\phi| < 1$.

```
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.arima_process import ArmaProcess

np.random.seed(42)

# statsmodels uses the lag-polynomial convention:
# AR: 1 - phi*B; MA: 1 + theta*B
phi = 0.6
theta = 0.4

ar = np.array([1, -phi])
ma = np.array([1, theta])

process = ArmaProcess(ar, ma)
y = process.generate_sample(nsample=500, burnin=200)

plt.figure(figsize=(11, 4))
plt.plot(y)
plt.title("Simulated ARMA(1,1) Time Series")
plt.xlabel("Time")
plt.ylabel("Y")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

6.3 Plotting the ACF and PACF

```
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 2, figsize=(12, 4))
plot_acf(y, lags=30, ax=axes[0])
plot_pacf(y, lags=30, method="ywm", ax=axes[1])
axes[0].set_title("Sample ACF")
axes[1].set_title("Sample PACF")
plt.tight_layout()
```

```
plt.show()
```

For an ARMA process, both ACF and PACF generally tail off. The plots are therefore used to propose a small collection of candidate models rather than to identify p and q mechanically.

6.4 Fitting a single ARMA model

In current versions of `statsmodels`, ARMA models are fitted using the `ARIMA` class with $d = 0$:

```
from statsmodels.tsa.arima.model import ARIMA

model = ARIMA(y, order=(1, 0, 1), trend="c")
result = model.fit()

print(result.summary())
print("Estimated parameters:")
print(result.params)
```

The order argument is (p, d, q). Setting $d = 0$ gives an ARMA model rather than a differenced ARIMA model. The constant/trend specification should be selected consistently with the intended mean parameterisation.

6.5 Fitting competing ARMA models

Fit several candidate orders to the same observations:

```
from statsmodels.tsa.arima.model import ARIMA
import pandas as pd

orders = [(1, 0, 0), (0, 0, 1), (1, 0, 1),
          (2, 0, 1), (1, 0, 2), (2, 0, 2)]

rows = []
fitted = {}

for order in orders:
    try:
        fit = ARIMA(y, order=order, trend="c").fit()
        fitted[order] = fit
        rows.append({
            "order": order,
            "AIC": fit.aic,
            "BIC": fit.bic,
            "logLik": fit.llf
        })
    except Exception as error:
        print(f"Model {order} failed: {error}")

comparison = pd.DataFrame(rows).sort_values("AIC")
print(comparison.to_string(index=False))
```

The model with the lowest AIC is the preferred model according to AIC among the candidates. Repeat the ranking using BIC if desired. Do not select a model solely because it has the smallest information criterion.

6.6 Residual diagnostics

```
best_order = comparison.iloc[0]["order"]
best_fit = fitted[best_order]

# Standard diagnostic plots
best_fit.plot_diagnostics(figsize=(12, 8))
plt.tight_layout()
plt.show()

# Ljung-Box test on residuals
from statsmodels.stats.diagnostic import acorr_ljungbox

residuals = best_fit.resid
lb = acorr_ljungbox(residuals, lags=[10, 15, 20],
                    model_df=best_order[0] + best_order[2],
                    return_df=True)
print(lb)
```

Interpret the Ljung–Box p-values at a preselected significance level, for example 0.05. If a p-value is small, there is evidence of remaining serial correlation at the tested lag range. Consider revising the model and repeating diagnostics.

6.7 Forecasting

```
forecast_steps = 12
forecast_result = best_fit.get_forecast(steps=forecast_steps)

forecast_mean = forecast_result.predicted_mean
forecast_ci = forecast_result.conf_int()

plt.figure(figsize=(11, 4))
plt.plot(y[-60:], label="Observed")
future_index = np.arange(len(y), len(y) + forecast_steps)
plt.plot(future_index, forecast_mean, label="Forecast")
plt.fill_between(
    future_index,
    forecast_ci.iloc[:, 0],
    forecast_ci.iloc[:, 1],
    alpha=0.2,
    label="95% prediction interval"
)
plt.title(f"Forecast from ARMA model {best_order}")
plt.xlabel("Time")
plt.ylabel("Value")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

Point forecasts are conditional expectations given the information available at the forecast origin. Beyond the first step, future innovations are unknown and are replaced by their conditional expectations, usually zero. Prediction intervals widen with the forecast horizon because uncertainty accumulates.

7 Worked Model Identification Example

Suppose a researcher observes a stationary monthly series and obtains the following qualitative correlogram features:

- The ACF declines gradually over several lags.
- The PACF has prominent spikes at lags 1 and 2, with smaller values thereafter.

An AR(2) model is a reasonable initial candidate because the PACF appears to cut off after lag 2. However, an ARMA$(1,1)$ or another low-order model may also fit the finite sample. The researcher should estimate the candidates, compare AIC/BIC, and inspect residual diagnostics before choosing.

7.1 Illustrative numerical ARMA calculation

Consider

$$Y_t - 50 = 0.5(Y_{t-1} - 50) + \varepsilon_t + 0.3\varepsilon_{t-1}.$$

If $Y_{t-1} = 54$, $\varepsilon_t = -1$ and $\varepsilon_{t-1} = 2$, then

$$Y_t = 50 + 0.5(54 - 50) - 1 + 0.3(2) = 50 + 2 - 1 + 0.6 = 51.6.$$

The autoregressive contribution is 2, while the net innovation contribution is -0.4.

8 Practical Considerations and Common Mistakes

1. **Fitting ARMA to a nonstationary series:** inspect the plot and stationarity evidence first. A trend or unit root may require transformation or differencing; differencing changes the model class to ARIMA.
2. **Overfitting:** high orders can absorb random noise. Begin with parsimonious candidates.
3. **Ignoring residual diagnostics:** a low AIC does not guarantee white-noise residuals.
4. **Confusing ACF and PACF:** for a pure AR model, PACF is the main order clue; for a pure MA model, ACF is the main order clue.
5. **Ignoring invertibility and causality:** check that the fitted solution is admissible and that the software has not returned a problematic parameterisation.
6. **Comparing incompatible likelihoods:** use the same observations, response transformation and likelihood conventions.
7. **Using in-sample fit as the only forecast criterion:** when forecasting is the goal, reserve a test period or use rolling-origin validation.

9 Summary

An ARMA(p,q) model combines autoregressive dependence on past observations with moving-average dependence on current and past innovations. The AR order is p and the MA order is q. Standard ARMA modelling assumes a stationary series and a well-behaved innovation process. Causality is determined by the roots of the AR polynomial, while invertibility is determined by the roots of the MA polynomial.

ACF and PACF plots provide initial clues about candidate orders. Parameters may be estimated by conditional least squares or maximum likelihood. Competing models are compared using AIC,

AICc and BIC, but the final model must also pass residual diagnostics. A good model should be parsimonious, leave approximately uncorrelated residuals and perform adequately for the intended forecasting task.

10 Tutorial and Assessment Exercises

Exercise 1: Model specification

For each of the following, write the full time-domain equation and identify the AR and MA orders:

(a) ARMA$(1, 1)$

(b) ARMA$(2, 1)$

(c) ARMA$(1, 2)$

(d) ARMA$(2, 2)$

Exercise 2: Numerical calculation

A time series follows

$$Y_t = 20 + 0.4(Y_{t-1} - 20) + 0.2(Y_{t-2} - 20) + \varepsilon_t + 0.5\varepsilon_{t-1}.$$

Given $Y_{t-1} = 24$, $Y_{t-2} = 22$, $\varepsilon_t = 1.5$ and $\varepsilon_{t-1} = -2$, calculate Y_t and interpret the contributions.

Exercise 3: Identification

Explain the expected ACF and PACF patterns for:

(a) AR(2);

(b) MA(1);

(c) ARMA$(1, 1)$.

Why are these patterns only guidelines when applied to sample data?

Exercise 4: Estimation

Explain the difference between conditional least squares and maximum likelihood estimation for ARMA models. Discuss one advantage and one limitation of each method.

Exercise 5: Model selection

Three candidate models have the following information criteria:

Model	AIC	BIC
ARMA$(1, 1)$	245.6	254.2
ARMA$(2, 1)$	243.1	255.0
ARMA$(2, 2)$	244.0	259.5

(a) Which model is preferred by AIC?

(b) Which model is preferred by BIC?

(c) Explain why the two criteria may select different models.

(d) What additional diagnostics should be considered?

Exercise 6: Python practical

Using a real or simulated stationary time series:

1. Plot the series.
2. Plot and interpret its ACF and PACF.
3. Fit at least four candidate ARMA models.
4. Prepare a table containing model order, log-likelihood, AIC and BIC.
5. Examine residual plots and conduct a Ljung–Box test.
6. Select a final model and justify your choice.
7. Produce forecasts for the next 12 periods and interpret the prediction intervals.

11 Suggested Structure for the Practical Report

Students should submit a concise report containing:

1. Description and source of the time series.
2. Initial time plot and stationarity assessment.
3. ACF/PACF plots and proposed candidate models.
4. Estimation method and parameter estimates.
5. Model comparison table with AIC and BIC.
6. Residual diagnostic plots and Ljung–Box results.
7. Final model selection with statistical justification.
8. Forecast plot and practical interpretation.